Xiaolin Zhou

+1 (323) 600-4608 | xzhou226@asu.edu | Homepage | Google Scholar | GitHub | ASU GLAD Lab

Education

Arizona State University — *Tempe, AZ*
Ph.D. in Computer Information Systems (Advisor: Prof. Xiyang Hu) — *Fall 2026 – Present*

University of Southern California — *Los Angeles, CA*
Master of Science in Computer Science (Advisor: Prof. Yue Zhao) — *Jan 2024 – Dec 2025*

Guangzhou University — *Guangzhou, China*
B.S. in Information and Computing Science — *Sep 2019 – Jul 2023*

Publications & Manuscripts

* denotes equal contribution

- **X. Zhou**, A. Yuan, Z. Luo, Z. Ling, X. Pan, Y. Gao, H. Zhang, J. Li, S. Jiang, P. Z. Wang, Z. Zhu, J. Liu, R. A. Rossi, H. Wei, and X. Hu. "When Simulation Lies: A Sim-to-Real Benchmark and Domain-Randomized RL Recipe for Tool-Use Agents." **NeurIPS 2026 ED Track**. arXiv:2605.11928.
- Y. Gao, **X. Zhou**, Y. Li, Y. Zhao, and R. Liu. "MedExAgent: Training LLM Agents to Ask, Examine, and Diagnose in Noisy Clinical Environments." **NeurIPS 2026**. arXiv:2605.07058.
- Y. Zhao, J. Li, L. Li, Y. Nian, J. Liu, **X. Zhou**, and X. Hu. "Auditable Claims about AI Agents." *Under Submission*, arXiv:2610.07459.
- P. Z. Wang*, C. Liang*, Z. Xu*, A. Yuan*, **X. Zhou***, H. Zhang, Y. Zhao, X. Hu, and S. Jiang. "CUA-SWE: When Computer-Use Agents Meet Visual Software Engineering." *Under Review*, arXiv:2609.32600.
- **X. Zhou**, J. Liu, L. Li, R. A. Rossi, and X. Hu. "Counterfactual Trace Auditing of LLM Agent Skills." *Under Review*, arXiv:2605.11946.
- **X. Zhou***, Z. Luo*, Y. Gao*, Q. Chen, X. Hu, Y. Zhao, and R. Liu. "Fairness or Fluency? An Investigation into Language Bias of Pairwise LLM-as-a-Judge." *Under Review*, arXiv:2601.13649.
- T. Zhang, **X. Zhou**, Y. Wang, E. Cambria, D. Traum, and R. Mao. "Individualized Cognitive Simulation in Large Language Models: Evaluating Different Cognitive Representation Methods." *Under Review*, arXiv:2510.20252.
- Z. Hu, **X. Zhou**, and M. Rostami. "Static Key Attention in Vision." arXiv:2412.07049, 2024.
- **X. Zhou**. "Eye-Blink Detection under Low-Light Conditions Based on Zero-DCE." *2022 IEEE Conference on Telecommunications, Optics and Computer Science (TOCS)*, pp. 1414–1417, 2022.
- **X. Zhou**. "Optimal Values Selection of Q-learning Parameters in Stochastic Mazes." *Journal of Physics: Conference Series*, vol. 2386, no. 1, p. 012037, IOP Publishing, 2022.

Research Experience

Research Collaborator (Co-first Author) — *May 2026 – Present*
Project lead: Prince Zizhuang Wang (CMU) — *Arizona State University*

- **CUA-SWE: Visual Software Engineering with Computer-Use Agents**
  - Designed and validated hybrid coding and computer-use tasks across Web, Game, DevOps, and Mobile for CUA-SWE, a 105-task benchmark, with deterministic verifiers and controlled task variants probing visual and runtime reasoning.
  - Analyzed task outcomes and agent trajectories to characterize GPT-6-Astra's advantage over evaluated frontier models, identifying post-edit GUI verification, accurate reconstruction of visual relationships, and runtime-guided revision as behaviors associated with stronger performance.

Research Assistant *Oct 2025 – May 2026*
Prof. Xiyang Hu *Arizona State University*

- **Sim-to-Real Robustness Benchmark & RL Recipe for Tool-Use Agents (NeurIPS 2026)**
  - Built **RobustBench-TC**, a tool-use robustness benchmark with 22 perturbation types organized along four POMDP components (observation, action, reward-relevant metadata, transition), each grounded in a verified GitHub issue or documented tool-calling failure.
  - Evaluated 21 models (1.5B–32B, including closed-source o4-mini) and exposed a sharply uneven robustness profile: observation perturbations cause $< 5\%$ accuracy drops, while reward-relevant and transition perturbations cause $\sim$40% and $\sim$30% drops, with model scale alone failing to close the gap.
  - Proposed **ToolRL-DR**, a domain-randomization RL recipe; on a 3B backbone it retains $\sim$75% of clean accuracy, matches open-source 14B function-calling baselines, and closes $\sim$27% of the unseen transition gap, indicating an emergent retry policy that transfers to runtime failures.

Research Assistant *Feb 2026 – May 2026*
Prof. Ruishan Liu & Prof. Yue Zhao *University of Southern California, FORTIS Lab*

- **MedExAgent: RL-Trained Clinical Diagnosis Agent (NeurIPS 2026)**
  - Formalized interactive clinical diagnosis as a POMDP with three action types (patient questioning, ordering exams as tool calls, issuing a diagnosis) and a systematic noise model spanning seven patient-noise and three exam-noise types.
  - Trained **MedExAgent** via a two-stage pipeline: SFT on Calgary–Cambridge-structured synthetic dialogues, then DAPO optimizing a composite reward over diagnostic accuracy, tool-call quality, and exam cost, matching larger models while keeping examinations cost-efficient.

Research Assistant *Feb 2026 – May 2026*
Prof. Xiyang Hu *Arizona State University*

- **Counterfactual Trace Auditing (CTA) of LLM Agent Skills**
  - Designed **CTA**, a framework that pairs with-skill and without-skill agent traces on the same task, segments them into goal-directed phases, aligns the phases, and emits structured Skill Influence Pattern (SIP) annotations measuring how a skill changes behavior rather than only its task outcome.
  - Instantiated CTA on SWE-Skills-Bench with Claude across 49 software-engineering tasks; surfaced an evaluation gap where mean pass rate shifts only +0.3 pp yet 522 SIP instances reveal substantial behavioral changes (template copying, off-task artifacts, excess planning, task recovery), turning informal failure-mode observations into reproducible behavioral measurements.

Professional Service

Program Committee Member, INFORMS Workshop on Data Science 2026
Reviewer, ACM Transactions on Intelligent Systems and Technology (TIST) 2026

Honors

- **First-Class Scholarship & Top 10 Students (0.01%)**, Guangzhou University 2021, 2022
- **National Second Prize**, China Undergraduate Mathematical Contest in Modeling 2020

Technical Skills

Programming: Python, Java, C++, JavaScript, SQL (MySQL)
ML / LLM: PyTorch, vLLM, RL post-training (GRPO, PPO, DPO, DAPO), SFT